

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277618

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Al-Harbi; Mazen S.

PROCESSES AND SYSTEMS FOR MONITORING AND CONTROLLING REFRIGERATION ENERGY COMSUMPTION

Abstract

Methods and systems include, using a monitoring system, initiating a control logic configured to control a production control system pertaining to a refrigeration system connected to a plurality of production trains. The method may include obtaining a plurality of production parameters; calculating, using the plurality of production parameters, an isentropic efficiency value for the primary compressor if the production flow rate is less than a production threshold; comparing each of the plurality of isentropic efficiency values; determining from the comparing each of the plurality of isentropic efficiency values, a primary compressor having a lowest isentropic efficiency value; selecting the respective primary compressor having the lowest isentropic efficiency value. The method may include, using a production control system, shutting down the selected primary compressor.

Inventors: Al-Harbi; Mazen S. (Al-Mubarraz, SA)

Applicant: SAUDI ARABIAN OIL COMPANY (Dhahran, SA)

Family ID: 96881045

Assignee: SAUDI ARABIAN OIL COMPANY (Dhahran, SA)

Appl. No.: 18/592074

Filed: February 29, 2024

Publication Classification

Int. Cl.: F25J1/02 (20060101)

U.S. Cl.:

CPC F25J1/0252 (20130101); F25J2230/24 (20130101); F25J2245/02 (20130101); F25J2280/50 (20130101); F25J2290/02 (20130101)

Background/Summary

BACKGROUND

[0001] Oil and gas wells produce fluids that flow from the wells to production plants. Production plants have facilities and systems to process the production fluids. Processing production fluids may include separation of production fluids into various forms such as water, oil, and gas. Water may be recycled and treated for fracking operations and/or may be disposed of in various ways. Oil may be piped to tankers for shipping. Gas may be piped to holding facilities but is unable to be shipped in a gas phase. Gas must be liquified for safe and efficient shipping. A refrigeration system is needed to chill the gas to an acceptable level for conversion to a liquified phase.

SUMMARY

[0002] This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

[0003] In some aspects, the techniques described herein relate to a method for monitoring and controlling energy consumption of a refrigeration system operatively connected to a plurality of production trains. The method may include initiating, using a monitoring system, a control logic configured to control a production control system pertaining to the refrigeration system. The plurality of production trains may include a production flow therethrough. Each of the production trains respectively may include a primary compressor including a first recycle valve. For each production train, the method may include obtaining, using the monitoring system, a plurality of production parameters. One of the plurality of production parameters may include a production flow rate. The method may include determining, using the monitoring system, whether the production flow rate is less than a production threshold. The method may include calculating, using the plurality of production parameters, an isentropic efficiency value for the primary compressor if the production flow rate is less than a production threshold. The method may include comparing, using the monitoring system, each of the plurality of isentropic efficiency values. The method may include determining, using the monitoring system, from the comparing each of the plurality of isentropic efficiency values, a primary compressor having a lowest isentropic efficiency value. The lowest isentropic efficiency value may be one of the plurality of isentropic efficiency values that is quantitatively smallest compared to all other isentropic efficiency values. The method may include selecting, using the monitoring system, the respective primary compressor having the lowest isentropic efficiency value. The method may include reducing, using the production control system, the production flow to the one of the plurality of production trains which includes the selected primary compressor. The method may include redirecting, using the production control system, the production flow from the one of the plurality of production trains including the selected primary compressor to at least one other of the plurality of production trains. The method may include opening, using the production control system, the first recycle valve of the selected primary compressor to fully opened. The method may include shutting down the selected primary compressor.

[0004] In some aspects, the techniques described herein relate to a system for monitoring and controlling energy consumption of a refrigeration system operatively connected to a plurality of production trains. The system may include a monitoring system configured to initiate a control logic configured to control a production control system pertaining to the refrigeration system. The plurality of production trains includes a production flow therethrough. Each of the production trains respectively includes a primary compressor including a first recycle valve. The monitoring system is configured to obtain a plurality of production parameters, wherein one of the plurality of production parameters includes a production flow rate. The monitoring system is configured to determine whether the production flow rate is less than a production threshold. The monitoring system is configured to calculate, using the plurality of production parameters, an isentropic efficiency value for the primary compressor if the production flow rate is less than a production threshold. The monitoring system is configured to compare each of the plurality of isentropic efficiency values. The monitoring system is configured to determine from the comparing each of the plurality of isentropic efficiency values, a primary compressor having a lowest isentropic efficiency value. The lowest isentropic efficiency value is one of the plurality of isentropic efficiency values that is quantitatively smallest compared to all other isentropic efficiency values. The monitoring system is configured to select the respective primary compressor having the lowest isentropic efficiency value. The production control system is configured to reduce the production flow to the one of the plurality of production trains which includes the selected primary compressor. The production control system is configured to redirect the production flow from the one of the plurality of production trains including the selected primary compressor to at least one other of the plurality of production trains. The production control system is configured to open the first recycle valve of the selected primary compressor to fully opened; and shut down the selected primary compressor.

[0005] Other aspects and advantages of the claimed subject matter will be apparent from the following description and the appended claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] Specific embodiments of the disclosed technology will now be described in detail with reference to the accompanying figures. Like elements in the various figures are denoted by like reference numerals for consistency.

[0007] FIG. 1 depicts a schematic diagram of a system in accordance with one or more embodiments.

[0008] FIG. 2 depicts a refrigeration system in accordance with one or more embodiments.

[0009] FIG. 3 depicts a flowchart in accordance with one or more embodiments.

[0010] FIG. 4 depicts a computer system in accordance with one or more embodiments.

DETAILED DESCRIPTION

[0011] In the following detailed description of embodiments of the disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art that the disclosure may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.

[0012] Throughout the application, ordinal numbers (e.g., first, second, third, etc.) may be used as an adjective for an element (i.e., any noun in the application). The use of ordinal numbers is not to imply or create any particular ordering of the elements nor to limit any element to being only a single element unless expressly disclosed, such as using the terms “before,” “after,” “single,” and other such terminology. Rather, the use of ordinal numbers is to distinguish between the elements.

By way of an example, a first element is distinct from a second element, and the first element may encompass more than one element and succeed (or precede) the second element in an ordering of elements.

[0013] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise.

[0014] Terms such as “approximately,” “substantially,” etc., mean that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

[0015] It is to be understood that one or more of the steps shown in the flowchart may be omitted, repeated, and/or performed in a different order than the order shown. Accordingly, the scope disclosed herein should not be considered limited to the specific arrangement of steps shown in the flowchart.

[0016] Although multiple dependent claims are not introduced, it would be apparent to one of ordinary skill that the subject matter of the dependent claims of one or more embodiments may be combined with other dependent claims.

[0017] In the following description of FIGS. **1-4**, any component described with regard to a figure, in various embodiments disclosed herein, may be equivalent to one or more like-named components described with regard to any other figure. For brevity, descriptions of these components will not be repeated with regard to each figure. Thus, each and every embodiment of the components of each figure is incorporated by reference and assumed to be optionally present within every other figure having one or more like-named components. Additionally, in accordance with various embodiments disclosed herein, any description of the components of a figure is to be interpreted as an optional embodiment which may be implemented in addition to, in conjunction with, or in place of the embodiments described with regard to a corresponding like-named component in any other figure.

[0018] While FIG. **1** shows various configurations of components, other configurations may be used without departing from the scope of the disclosure. For example, various components in FIG. **1** may be combined to create a single component. As another example, the functionality performed by a single component may be performed by two or more components.

[0019] Turning to FIG. **1**, FIG. **1** shows a schematic diagram in accordance with one or more embodiments. As shown in FIG. **1**, a system for monitoring and controlling energy consumption (hereafter “energy consumption system”) (**10**) may include a gas production network (**100**), a user device (**150**), a monitoring system (**120**), and various network elements (not shown). In some embodiments, the gas production network (**100**) may include a gas well (**101**), a gathering system (**103**), and a gas plant (**105**). In one or more embodiments, the monitoring system (**120**) may be a gas plant server. In some embodiments, various types of gas production data (**130**) are collected over the energy consumption system (**10**). Likewise, the energy consumption system (**10**) may also obtain compressor status data (**136**) and/or power consumption data (**142**) regarding one or more compressors throughout the energy consumption system (**10**). The compressors may be configured to compress gas, refrigerant or other fluids from processing production fluids.

[0020] Furthermore, the gas well (**101**) may include a well system (**102**) located in a well environment that includes a hydrocarbon reservoir (“reservoir”) located in a subsurface hydrocarbon-bearing formation. The hydrocarbon-bearing formation may include a porous or fractured rock formation that resides underground, beneath the earth's surface (“surface”). In the case of the well system (**102**) being a hydrocarbon well, the reservoir may include a portion of the hydrocarbon-bearing formation. The hydrocarbon-bearing formation and the reservoir may include different layers of rock having varying characteristics, such as varying degrees of permeability, porosity, and resistivity. In the case of the well system (**102**) being operated as a production well,

the well system (102) may facilitate the extraction of hydrocarbons (or “production”) from the reservoir. In some embodiments, the well system (102) includes a wellbore, a well sub-surface system, a well surface system, and a well control system. The wellbore may include a bored hole that extends from the surface into a target zone of the hydrocarbon-bearing formation, such as the reservoir. The wellbore may facilitate the circulation of drilling fluids during drilling operations, the flow of hydrocarbon production (“production”) (e.g., oil and gas) from the reservoir to the surface during production operations, the injection of substances (e.g., water) into the hydrocarbon-bearing formation or the reservoir during injection operations, or the communication of monitoring devices (e.g., logging tools) into the hydrocarbon-bearing formation or the reservoir during monitoring operations (e.g., during in situ logging operations). A well control system in a well system (102) may control various operations of the well system (102), such as well production operations, well completion operations, well maintenance operations, and reservoir monitoring, assessment and development operations. In some embodiments, the well control system includes a computer system (400) that is the same as or similar to that of computer (402) described below in FIG. 4 and the accompanying description.

[0021] In some embodiments, one or more gas wells are coupled to the gathering system (103). The gathering system (also referred to as a collecting system or gathering facility) (103) may include various hardware arrangements and pipe components that connect gas flowlines from several gas wells into a single gathering line. For example, a gathering system may include flowline networks, headers, pumping facilities, separators, emulsion treaters, compressors, dehydrators, tanks, valves, regulators, and/or associated equipment. In particular, a production header (104) may have production valves and testing valves to control a mixed stream for a flowline of a respective gas well. Thus, a gathering system may direct various hydrocarbon fluids to a processing or testing facility, such as a gas plant. In some embodiments, a gathering system manages individual fluid ratios (e.g., a particular gas-to-water ratio or condensate-to-gas ratio) as well as supply rates of oil, gas, and water. For example, a gathering system may assign a particular production value or ratio value to a particular gas well by opening and closing selected valves among the production headers and using individual metering equipment or separators. Furthermore, a gathering system may be a radial system or a trunk line system. A radial system may bring various flowlines to a single central header. In contrast, a trunk-line system may use several production headers to collect oil and gas from fields that cover a large geographic area. Once collected, a gathering system may transport and control the flow of oil or gas to a storage facility, a gas processing plant, or a shipping point.

[0022] Keeping with FIG. 1, gas may be transported from one or more gas wells to one or more gas plants. More specifically, a gas plant may refer to various types of industrial plants such as a gas processing plant, a gas cycling plant, or a compressor plant. A gas processing plant (also referred to as a natural gas processing plant) may be a facility that processes natural gas to recover natural gas liquids (e.g., condensate, natural gasoline, and liquefied petroleum gas) and sometimes other substances such as sulfur. A gas cycling plant may refer to an oilfield installation coupled to a gas-condensate reservoir. In particular, a gas cycling plant may extract various liquids from natural gas. Consequently, the remaining dry gas may be compressed prior to returning to a producing formation, e.g., to maintain reservoir pressure. Moreover, various components of natural gas may be classified according to their vapor pressures, such as low pressure liquid (i.e., condensate), intermediate pressure liquid (i.e., natural gasoline), and high pressure liquid (i.e., liquefied petroleum gas). Examples of natural gas liquids include propane, butane, pentane, hexane, and heptane. With respect to compressor plants, a compressor plant may be a facility that includes multiple gas compressors, auxiliary treatment equipment, and pipeline installations for pumping natural gas over long distances. A compressor station may also repressurize gas in large gas pipelines or to link offshore gas fields to their final terminals.

[0023] Keeping with gas plants, the gas plant (105) may include water processing equipment that includes hardware and/or software for extracting, treating, and/or disposing of water associated

with gas processing. More specifically, the gas plant (**105**) may extract produced water during the separation of oil or gas from a mixed fluid stream acquired from a gas well. This produced water may be a kind of brackish and saline water brought to the surface from underground formations. In particular, oil and gas reservoirs may have water in addition to hydrocarbons in various zones underneath the hydrocarbons, and even in the same zone as the oil and gas. However, most produced water is of very poor quality and may include high levels of natural salts and minerals that have dissociated from geological formations in the target reservoir. Likewise, produced water may also acquire dissolved constituents from fracturing fluids (e.g., substances added to the fracturing fluid to help prevent pipe corrosion, minimize friction, and aid the fracking process). However, through various water treatments, produced water may be reused in one or more gas wells, e.g., through waterflooding where produced water is injected into the reservoirs. By injecting produced water into an injection well, the injected water may force oil and gas to one or more production wells.

[0024] Keeping with produced water, the gas plant (**105**) may use various treatment technologies in order to reuse or dispose of produced water, such as conventional treatments and advanced treatments. For example, conventional treatments may include flocculation, coagulation, sedimentation, filtration, and lime softening water treatment processes. Thus, conventional treatment processes may include functionality for removing suspended solids, oil and grease, hardness compounds, and other insoluble water components. With advanced treatment technologies, water processing equipment may include functionality for performing reverse osmosis membranes, thermal distillation, evaporation and/or crystallization processes. These advanced treatment technologies may treat dissolved solids, such as chlorides, salts, barium, strontium, and sometimes dissolved radionuclides. In some embodiments, produced water is sent to a wastewater treatment plant that is equipped to remove barium and strontium, e.g., using sulfate precipitation. Outside of treatments for reusing produced water, water processing equipment may dispose of produced water using various water management options. For example, produced water may be disposed of in saltwater wells. Likewise, produced water may also be eliminated through a deep well injection.

[0025] In some embodiments, the gas plant (**105**) may include one or more pipe components, one or more storage facilities, and one or more production control systems. For example, different forms of gas may be stored in various storage facilities that include surface containers as well as various underground reservoirs, such as depleted gas reservoirs, aquifer reservoirs and salt cavern reservoirs. With respect to production control systems, a production control system (**107**) may include hardware and/or software that monitors and/or operates equipment, such as at a gas well or in a gas plant. Examples of production control systems may include one or more of the following: an emergency shut down (ESD) system, a safety control system, a video management system (VMS), process analyzers, other industrial systems, etc. In particular, the production control system (**107**) may include a programmable logic controller that may control valve states, fluid levels, pipe pressures, warning alarms, pressure releases and/or various hardware components for implementing a gas flowline. Thus, a programmable logic controller may be a ruggedized computer system with functionality to withstand vibrations, extreme temperatures, wet conditions, and/or dusty conditions, such as those around a gas plant, gas well, and/or a gathering system.

[0026] With respect to distributed control systems, a distributed control system may be a computer system for managing various processes at a facility using multiple control loops. As such, a distributed control system may include various autonomous controllers (such as remote terminal units (RTUs)) positioned at different locations throughout the facility to manage operations and monitor processes. Likewise, a distributed control system may include no single centralized computer for managing control loops and other operations. On the other hand, a SCADA system may include a control system that includes functionality for enabling monitoring and issuing of process commands through local control at a facility as well as remote control outside the facility.

With respect to an RTU, an RTU may include hardware and/or software, such as a microprocessor, that connects sensors and/or actuators using network connections to perform various processes in the automation system.

[0027] Keeping with production control systems, the production control system (107) may be coupled to facility equipment. Facility equipment may include various machinery such as one or more hardware components, such as pipe components, refrigeration system components and/or electrical system components, that may be monitored using one or more sensors. Examples of hardware components coupled to the production control system (107) may include crude oil preheaters, heat exchangers, pumps, valves, compressors, loading racks, and storage tanks among various other types of hardware components. Hardware components may also include various network elements or control elements for implementing control systems, such as switches, routers, hubs, PLCs, remote terminal units, user equipment, or any other technical components for performing specialized processes. Examples of sensors may include pressure sensors, flow rate sensors, temperature sensors, torque sensors, rotary switches, weight sensors, position sensors, microswitches, hydrophones, accelerometers, etc. Monitoring systems, user devices, and network elements may be computer systems similar to the computer system (400) described in FIG. 4 and the accompanying description.

[0028] In some embodiments, gas may be processed in the gas plant (105) through one or more production trains. A production train (112) may be configured to liquefy natural gas for gas transport. The production train (112) may include a production flow therethrough. Production flow may include the flow of production fluids such as condensate, gas and/or water. The production train (112) may include a refrigeration system (106). The refrigeration system (106) may be configured to cool and condense gas from one or more gas wells. The production train (112) may be a facility that includes multiple refrigeration equipment, and pipeline installation for liquefying natural gas for transport such as shipping. The refrigeration system (106) may include refrigeration components such as one or more refrigerant compressors (e.g., a primary refrigerant compressor (108) and/or a secondary refrigerant compressor (109)), a chiller (208), and a refrigerant tank (212). Refrigeration components may also include piping, one or more sensors, and control elements coupled to the production control system (107). The refrigerant compressors may include hardware components such as recycle valves (e.g., a first recycle valve (110) and/or a second recycle valve (111)), motors, tubing, fittings, and the like.

[0029] In some embodiments, the monitoring system (120) may include hardware and/or software for collecting data in real-time from various gas wells, gas plants, sensors coupled to hardware equipment and pipe components, user devices, and other systems (e.g., electrical, alarm) in the energy consumption system (10). For example, the monitoring system (120) may be one or more plant servers with functionality for obtaining data throughout the energy consumption system (10), such as gas production data (130), refrigeration system data (139) (e.g., compressor status data (136)). For example, gas production data (130) may include operating upstream and downstream sensor data for various pipe components (e.g., pressure data, temperature measurements, and gas flow rates), production flow rates, and power consumption data (142) from various pipeline information (PI) systems, such as production control systems located throughout the energy consumption system (10). Gas production data (130) may also include gas chemical composition data, such as condensate-gas ratio (CGR) data, and water sampling data (e.g., levels of Chloride and Strontium concentrations). Likewise, gas production data (130) may also include material and design specifications for various gas production components that form gas flowlines, such as pipe component geometry and pipe component compositions. The monitoring system (120) may also collect various production parameters regarding gas plant operations, gas well operations, and production header information regarding the gathering system (103) coupled to the gas wells.

[0030] In some embodiments, the monitoring system (120) may include a power panel (132) configured to obtain various production parameters such as current, voltage, and power factor, and

the like. The power panel (132) may be operatively connected to electrical systems of the power plant and/or production trains. The power panel (132) may include hardware equipment and/or software for measuring current, voltage and other electrical parameters. Examples of hardware equipment for the power panel (132) may include, but is not limited to, voltmeters and/or ammeters.

[0031] In some embodiments, the monitoring system (120) includes functionality for determining and/or implementing one or more remediation operations based on integrity assessments, power consumption data (142). A remediation operation may include replacing a particular refrigeration system component that is part of a gas flowline based on the pipe component failing to satisfy a predetermined criterion (e.g., pipe thickness falling below an integrity threshold). Likewise, a remediation operation may also include adjusting gas production operations to manage corrosion levels in the corresponding refrigeration component. Likewise, a remediation operation may also include applying one or more lubricants to a particular refrigeration component to prevent future failure. In some embodiments, the monitoring system (120) may automatically prioritize various remediation procedures among different refrigeration components (e.g., refrigerant compressors, and heat exchangers) instantaneously based on desired gas production targets, future plant operations, and/or the corrosion states of various gas production network components.

[0032] In some embodiments, the monitoring system (120) may include a user interface (151). The user interface (151) may be integrated with the user device (150) or may be integrated with various components of the monitoring system (120) such as the system display (138). The user device (150) may communicate with the monitoring system (120) to present status reports to a particular user. Based on the status reports, the user device (150) may also manage various commands for performing one or more remediation operations based on one or more user selections. The user device (150) may be a personal computer, a handheld computer device such as a smartphone or personal digital assistant, or a human-machine interface (HMI). For example, a user may interact with the user interface (e.g., a graphical user interface) (151) to inquire regarding production network status and integrity levels in one or more gas production components at the gas plant (105). Through user selections or automation, the monitoring system (120) may identify compressors (e.g., refrigerant compressors) that fail power consumption criteria and implement remediation operations accordingly such as shutting down the refrigerant compressor.

[0033] In some embodiments, an integrity assessment (127) of one or more compressors is generated by the monitoring system (120) upon obtaining a request (e.g., request for a control logic initiation) from the user device (150) and using various predetermined criteria (e.g., power consumption criteria (142) and input data (e.g., gas production data (130)). The request may be a network message transmitted between the user device (150) and the monitoring system (120) that identifies a particular pipe component, gas production network system, or portion of the energy consumption system (10) for an operational analysis. In some embodiments, the monitoring system (120) includes functionality for transmitting commands to one or more production control systems to implement a particular remediation operation. For example, the monitoring system (120) may transmit a network message over a machine-to-machine protocol (e.g., a control logic) to the production control system (107) in the gas plant (105). A command may be transmitted periodically, based on a user input, or automatically based on changes in gas production data (130).

[0034] Returning to FIG. 1, the monitoring system (120) may include hardware and/or software with functionality for generating and/or updating one or more machine-learning models to determine predicted power consumption data (e.g., power consumption, and/or power load) (142), and/or predicted compressor assessment (e.g., by classifying a compressor as passing or failing one or more power consumption criteria). Examples of machine-learning models may include random forest models and artificial neural networks, such as convolutional neural networks, deep neural networks, and recurrent neural networks. Machine-learning models may also include support vector machines, decision trees, inductive learning models, deductive learning models, supervised

learning models, unsupervised learning models, reinforcement learning models, etc. In a deep neural network, for example, a layer of neurons may be trained on a predetermined list of features based on the previous network layer's output. Thus, as data progresses through the deep neural network, more complex features may be identified within the data by neurons in later layers. Likewise, a U-net model or other type of convolutional neural network model may include various convolutional layers, pooling layers, fully connected layers, and/or normalization layers to produce a particular type of output. Thus, convolution and pooling functions may be the activation functions within a convolutional neural network. In some embodiments, two or more different types of machine-learning models are integrated into a single machine-learning architecture, e.g., a machine-learning model may include K-nearest neighbor (k-NN) models and neural networks. In some embodiments, a control system simulator may generate augmented data or synthetic data to produce a large amount of modeled system data for training a particular model.

[0035] While FIG. 1 shows various configurations of components, other configurations may be used without departing from the scope of the disclosure. For example, various components in FIG. 1 may be combined to create a single component. As another example, the functionality performed by a single component may be performed by two or more components.

[0036] FIG. 2 illustrates an example embodiment of the refrigeration system (106), having a production side and a refrigerant side. The production fluid flows into (250) and out (255) of the energy consumption system (10) on the production side of the refrigeration system (106). The production fluid is cooled by refrigerant flowing on the refrigerant side of the refrigeration system (106).

[0037] The refrigeration system (106) includes a refrigerant circulation system (200), which may be an open loop or a closed loop system. The refrigerant circulation system (200) includes a refrigerant feed pipe fluidly connected to and configured to receive a flow (215) of refrigerant from an outlet of a refrigerant tank (212) to an inlet of a chiller (208). The refrigerant tank (212) is configured to store a refrigerant (213). A refrigerant effluent pipe fluidly connects to and configured to receive the flow (215) of refrigerant from a refrigerant-side outlet of the chiller (208) to an inlet of a refrigerant compressor (e.g., a primary refrigerant compressor (108) and/or a secondary compressor (109)). The refrigerant compressor is configured to compress the refrigerant (213). The compressor may be electrically coupled to the production control system (107). The production control system (107) may receive a request from the monitoring system (120) to start or shut down any compressor (e.g., a refrigerant compressor). The refrigerant compressor may include a recycle valve configured to control a quantity of inflowing gas to be recycled. Even though FIG. 2 only shows one refrigerant compressor, it will be obvious to someone skilled in the art that the refrigeration system (106) may include more than one refrigerant compressor.

[0038] In some embodiments, the refrigeration system (106) may include a primary refrigerant compressor (108) and a secondary refrigerant compressor (109) coupled to the refrigerant circulation system (200) and configured to receive the flow (215) of refrigerant from the chiller (208). In some embodiments, the refrigeration system (106) may include a refrigerant compressor scrubber (216). The inlet of the refrigerant compressor scrubber may be fluidly connected to the outlet of the chiller. The refrigerant compressor scrubber (216) is configured to remove particulates and debris from the refrigerant (213) before flowing into the refrigerant compressor. The outlet of the refrigerant compressor scrubber may be fluidly connected to the inlet of the refrigerant compressor.

[0039] Keeping with chillers, the chiller (208) may be configured to further cool or chill the compressed gas into a liquid form. The chiller (208) may include a refrigerant side and a gas side. The gas side of the chiller (208) may be fluidly connected to the gas plant (105) to receive the flow of natural gas after being processed in the gas plant (105) via one or more pipe components. The relatively warm gas flows in the production-side inlet of the chiller fluidly connected to the gas plant (105) by a gas pipeline configured to receive the flow of natural gas. The relatively colder gas

flows out the production-side outlet of the chiller by a gas pipeline configured to receive the flow of natural gas. The natural gas may be in a liquefied form. The liquefied natural gas may flow to a holding facility to be transported from the gas plant (105). The refrigeration system (106) may also include an expansion valve (210). The expansion valve (210) is configured to regulate the quantity of refrigerant (213) that is input into the chiller (208). The expansion valve (210) may be fluidly connected to the inlet of the chiller in accordance with one or more embodiments. The chiller (208) may include hardware components such as one or more chiller sensors (220) to provide feedback to the expansion valve (210). The chiller sensor (220) is coupled to the expansion valve (210). Examples of the chiller sensor may be a sensing bulb.

[0040] In some embodiments, the refrigeration system (106) further includes a heat exchanger (214) such as a condenser. The heat exchanger (214) is configured to remove heat from the refrigerant (213) thereby cooling the refrigerant (213) before being stored in the refrigerant tank (212). The heat exchanger (214) may include hardware components such as fittings, and tubing configured to control flow (215) of refrigerant. The inlet of the heat exchanger may be fluidly connected to the outlet of the refrigerant compressor. The outlet of the heat exchanger may be fluidly connected to the inlet of the refrigerant tank (212).

[0041] The refrigeration system (106) further includes one or more sensors configured to measure a property of the refrigerant (213) circulating in the refrigerant circulation system (200). For example, one or more temperature sensors may be disposed upstream and/or downstream of the heat exchanger (214). As another example, one or more flow rate sensors may be disposed upstream and/or downstream of the heat exchanger (214). As yet another example, one or more pressure sensors may be disposed upstream and/or downstream of the heat exchanger (214). Some embodiments may include combinations of these sensors upstream and/or downstream of the heat exchanger (214).

[0042] The production control system (107) also includes a digital control system. The digital control system may receive a signal from the one or more sensors, such as a signal from a temperature sensor, a signal from a pressure sensor, or a signal from a flow rate sensor. The digital control system may be configured to convert the signals, which may be provided in volts or amps, for example, to a measured unit, such as ° C., kg/h, barg, or other units commonly used for temperature, volume or mass flow, or pressure, among other properties that may be measured, and to display the measurement to an operator via a display device, for example.

[0043] The monitoring system (120) of embodiments herein is configured to provide an alert, such as an audible or visual alarm, when a signal (measurement) from at least one of the one or more sensors is indicative of power consumption overages and/or of an active leak from the process side of the heat exchanger (214) to the refrigerant side of the heat exchanger (214). Leaks may cause power consumption to increase as refrigerant compressors may be required to increase refrigerant flow rates through the refrigerant circulation system (200).

[0044] For example, the one or more sensors may include a temperature sensor configured to measure a temperature of the refrigerant (213) in the refrigerant circulation system (200). The digital control system may be configured to provide an alert when a temperature indicated by the signal from the temperature sensor is indicative of power consumption overages. Overuse of refrigerant compressors may result in excess carbon emissions and/or result in an increase in compressor failure.

[0045] As another example, the one or more sensors may include a pressure sensor configured to measure power consumption of the refrigerant compressors in the refrigerant system (106). The digital control system may be configured to provide an alert when power consumption indicated by the signal from the sensor is indicative of power consumption overages.

[0046] As yet another example, the one or more sensors may include a flow rate sensor configured to measure a flow rate of the refrigerant (213) in the refrigerant circulation system (200). The digital control system may be configured to provide an alert when a flow rate indicated by the

signal from the flow rate sensor is indicative of a leak from the process side of the heat exchanger (214) to the refrigerant side of the heat exchanger (214). Addition of process fluid to the refrigerant flow (215) may result in an increase or spike in the measured flow rate of the refrigerant (213), indicating a leak has occurred. Some embodiments may include combinations of these various sensors and associated alerts.

[0047] Continuous chemical processes generally operate at a “steady state,” other than during startup, shutdown, or transitions. During steady state operations, flows, temperatures and pressures will fluctuate to a minor degree. For example, steady state temperatures may include fluctuations in temperature by a few tenths of a degree to as much as a few degrees around a set point. Similarly, pressure and flow rates may fluctuate around a set point, the production control system (107) being used to control valves, pumps, and other equipment in order to maintain the set points and steady operations. Often, the process side of the operations is of primary concern, as that is the side that controls product quality and throughput. Fluctuations on the utility or refrigerant side are of lesser concern, and fewer instrumentation and sensors are used; as long as the process side is operating properly, it is generally of less concern as to how much the refrigerant side fluctuates.

[0048] In contrast, embodiments herein utilize sensors on the refrigerant side to detect leaks of corrosive fluids, so as to be able to quickly act to avoid corrosion throughout the refrigeration system (106). Alarms or alerts may be provided where the temperature, pressure, or flow is indicative of a leak into the refrigerant circulation system (200), the fluctuation in one or more of these values being greater than what is typical for steady state operations. An alarm (125) may alert when power consumption exceeds a power threshold. The atypical values may be a percentage or a delta from the steady state set point, and may be 5%, 7.5%, 10%, above/below the steady state set point, or may be a few degrees, a few bar, or a few kg/h (or other flow measurement unit) above/below the steady state set point, which when occurring, may result in an audible or visual alarm provided by the monitoring system (120). In other embodiments, the atypical value may be a delta value, such as a difference between inlet and outlet flow, inlet and outlet temperature, or inlet and outlet pressure. For example, a refrigerant may be provided to an inlet at a given pressure or temperature, with an expected pressure drop or temperature rise occurring across the exchanger, resulting in an outlet pressure or temperature. The (in-out) value may fluctuate over a typical range during “steady” operations; however, the difference between in and out and the typical delta may go outside the expected range when a leak occurs. In other words, alarms may be configured based on a single measured value according to some embodiments, or alarms may be configured based on a combination of measured values according to other embodiments. Operational alarms (investigative) may be provided in addition to the leak alarms (action required) and/or power consumption alarms, and may be set at a lesser value or delta closer to the typical steady state range of operations.

[0049] The audible or visual alarms provided may indicate to an operator that a leak is occurring or power thresholds have been exceeded, and that remedial action needs to be taken. For example, following the alert, a heat exchanger and/or a refrigerant compressor may be isolated, taken out of service, and inspected to find the leak(s) and/or energy efficiency issues. The leak(s) and/or efficiency issues may then be fixed, such as by replacing tube sheets or tube bundles or the entire exchanger, or the leak may be bypassed, such as by plugging an inlet end of a leaking tube, among other possible actions that could prevent further flow of process fluid into the refrigerant side of the exchanger. Examples of energy efficiency issues may be fixed by replacing compressor motors, clogged tubing and/or fittings. Isolation of a heat exchanger and/or refrigerant compressors may be performed manually by operators. In some embodiments, however, the production control system (107) may be configured to isolate a heat exchanger and/or compressors automatically when the signal is indicative of a leak and/or power consumption overages. As it may be undesired to take an exchanger and/or compressor out of service, as this may impact upstream and downstream operations, parallel heat exchangers and/or compressors may be provided, and the production

control system (107) may be configured to place a parallel heat exchanger and/or compressor in service while taking the leaking exchanger out of service, opening and closing respective valves so as to effect the transition of operations between the parallel heat exchangers and/or compressors. Where multiple parallel exchangers and/or compressors are provided and are used concurrently, the production control system (107) may be configured to take one or more of the leaking parallel exchangers and/or compressors out of service and to transition the process operations to reduced rates in view of the resulting reduced heat exchange capacity.

[0050] Embodiments herein thus provide for additional sensors, not typically used or considered during plant design and construction, so as to provide preventative measures during operation of the plant. Embodiments herein further provide for additional digital control system configurations to provide the alarms or alerts. The configuration of the sensors, controls, and control systems may be system dependent, and may depend on numerous factors, as may be appreciated by one skilled in the art. The additional capital expense of these sensors and controls is well valued, however, in view of the damage that may be avoided, and the associated capital saved following a leak event.

[0051] FIG. 3 depicts a flowchart in accordance with one or more embodiments describing a method for controlling refrigeration energy consumption (hereafter “control method”) (300). In some embodiments, the control method (300) may include using control systems and monitoring systems to control refrigeration energy consumption. Although the steps in flowchart using the control method (300) are shown in sequential order, it will be apparent to one of ordinary skill in the art that some steps may be conducted in parallel, in a different order than shown, or may be omitted without departing from the scope of the invention.

[0052] In step (302), the control method (300) may include initiating a control logic (135) pertaining to the refrigeration system (106) in accordance with one or more embodiments. Each production train may include production flow therethrough. Each production train includes a refrigerant compressor (e.g., a primary refrigerant compressor (108)). The refrigerant compressor may include a recycle valve. The monitoring system (120) may be configured to initiate the control logic (135). The control logic (135) is configured to control the production control system (107). The monitoring system (120) and production control system (107) may be operatively connected and configured to communicate with each other and the various components of the energy consumption system (10). The monitoring systems and controls systems may include computer systems that include at least the computer system (400) the same or similar to the computer (402) described in FIG. 4 and accompanying description. The computer (402) may include software programs to digitally run the control logic (135). The control logic (135) may be initiated from input entered on the user interface (151). The monitoring system (120) is configured to initiate the control logic (135) from input from the user interface (151).

[0053] In step (303), the control method (300) may include obtaining one or more production parameters for each production train in accordance with one or more embodiments. The one or more production parameters may include a production flow rate, a recycle valve opening measurement, a cooling duty, a driver power value, and/or electrical parameters such as current and voltage. The monitoring system (120) may be configured to obtain the production parameters.

[0054] In step (304), the control method (300) may include determining whether the production flow rate is less than a production threshold in accordance with one or more embodiments. The production threshold may be a numerical constant (e.g., 1450 million standard cubic feet per day (mmscfd) or a range). The production threshold may be entered by a user using the user interface (151) of the monitoring system (120) and/or the user device (150). The monitoring system (120) is configured to determine if the production flow rate is less than the production threshold. If the production flow rate is greater than the production threshold, then it may be determined that the compressor energy consumption may be within normal operating limits and no change is necessary. If the production flow rate is determined to be less than the production threshold, then an isentropic efficiency value may be calculated for each compressor (e.g., each primary compressor and each

secondary compressor).

[0055] In step (305), the control method may include calculating the isentropic efficiency value for each compressor using the one or more production parameters in accordance with one or more embodiments. The isentropic efficiency value may be calculated as follows:

$$[00001] \text{ eff} = \frac{(\text{totalflow} - \text{recycleflow})}{\text{totalflow}} \times \frac{H_{\text{isn}}}{\text{DriverPower}}, \quad (1)$$

with eff as the isentropic efficiency value, total flow as the total flow of refrigerant through the inlet of the refrigerant compressor, recycle flow is the flow (215) of refrigerant through the recycle valve, $\Delta H_{\text{sub.isn}}$ is the change in enthalpy at the inlet and outlet of the compressor given by the temperature of the fluid at the inlet and outlet of the compressor, and driver power is the power required to power the compressor. The driver power may be calculated as follows:

$$[00002] \text{ driverpower} = \sqrt{3} \times V \times I \times \text{PF}, \quad (2)$$

with V for voltage, I for current, and PF for power factor of the compressor motor given by the ratio of measured real power to measured apparent power. The monitoring system (120) may be configured to measure electrical parameters such as, but is not limited to, voltage, current, real power, and apparent power. The monitoring system (120) may be configured to determine the driver power from a measured voltage, a measured current, and a power factor. The monitoring system (120) may also be configured to calculate the isentropic efficiency value using the plurality of production parameters if the production flow rate is less than the production threshold. In some embodiments, the recycle flow may be calculated using refrigerant fluid parameters and/or valve specifications such as the recycle valve opening measurements, and/or valve flow coefficient. Examples of refrigerant fluid parameters are refrigerant specific gravity and/or refrigerant density. The valve specifications and refrigerant fluid parameters may be stored in the monitoring system (120) or entered using the user interface (151).

[0056] In step (306), the control method (300) may include comparing the isentropic efficiency value of each compressor in accordance with one or more embodiments. The monitoring system (120) may be configured to compare all of the isentropic efficiency values. In step (307), the control method (300) may include determining the lowest isentropic efficiency value from comparing all of the isentropic efficiency values. The lowest isentropic efficiency value is one of the isentropic efficiency values that is quantitatively smallest compared to all other isentropic efficiency values. The monitoring system (120) may be configured to determine from the one or more isentropic efficiency values the lowest isentropic efficiency value and the corresponding compressor (e.g., the primary compressor and/or secondary compressor) and the corresponding production train (112) of the compressor with the lowest isentropic efficiency value.

[0057] In step (308), the control method (300) may include selecting the respective compressor (e.g., the primary compressor) corresponding to the lowest isentropic efficiency value in accordance with one or more embodiments. The monitoring system (120) may be configured to select the compressor having the lowest isentropic efficiency value.

[0058] In step (310), the control method may include reducing, using the production control system (107), the production flow to the corresponding production train (112) which includes the selected compressor (e.g., selected primary compressor) in accordance with one or more embodiments. In some embodiments, if the compressor is the only compressor supporting the production train (112), the production flow may be shut off to the production train (112). The production control system (107) may be configured to reduce, shut down, and/or control the production flow to the production train (112) which includes the selected compressor. The production control system (107) may include hardware equipment to reduce, shut down and/or control the production flow to the production train (112).

[0059] In step (312), the control method (300) may include redirecting, using the production control system (107), the production flow from the corresponding production train (112) with the selected compressor to any other production train in accordance with one or more embodiments.

The production control system (107) may be configured to redirect flow from one of the production trains to other production trains. In some embodiments, the production flow is redirected to other production trains and redistributed equally among the other production trains that are not the production train having the lowest isentropic value.

[0060] In step (314), the control method (300) may include opening, at least partially, the recycle valve (e.g., the first recycle valve (110)) of the selected compressor in accordance with one or more embodiments. The production control system (107) may be configured to open the recycle valve of the selected compressor. In some embodiments, the recycle valve of the selected compressor may be fully opened thereby allowing unrestricted production flow through the recycle valve.

[0061] In step (316), the control method (300) may include shutting down the selected compressor (e.g., the primary compressor) corresponding to the lowest isentropic efficiency value in accordance with one or more embodiments. The production control system (107) may include hardware equipment for shutting down compressors. The compressor may be shut down by a signal from the monitoring system (120). The production control system (107) is configured to receive the shutdown signal from the monitoring system (120). The production control system (107) is also configured to shut down the compressor. In some embodiments, the recycle valve of the selected compressor may be, at least partially, opened so the refrigerant flow (215) is, at least partially, recycled before shutting down the selected compressor. In some embodiments, the recycle valve may be fully opened to allow full recycling of the refrigerant (213) through the compressor. The production control system (107) is configured to open the recycle valve.

[0062] In some embodiments, the control method (300) may include calculating the cooling duty of the production network needed for liquifying the production fluids. The cooling duty may be calculated as follow:

$$[00003] \text{ coolingduty} = (V \times \rho_{\text{fl}}) \times C_p \times \Delta T, \quad (3)$$

where V is production flow rate, $\rho_{\text{sub.f1}}$ is production fluid density, $C_{\text{sub.p}}$ is specific gravity of the production fluid, and ΔT is change in temperature of the production fluid at the chiller inlet and the temperature of the production fluid at the chiller outlet. The monitoring system (120) may be configured to obtain cooling duty parameters such as production fluid density, and/or specific gravity. The monitoring system (120) may also be configured to calculate the cooling duty.

[0063] In some embodiments, the control method (300) may include comparing the cooling duty to a cooling threshold. The cooling threshold may be entered by a user using the user interface (151) of the monitoring system (120) and/or the user device (150). If the cooling duty is greater than or equal to the cooling threshold, then normal operations may continue. If the cooling duty is less than the cooling threshold, then the control method may proceed until the selected compressor is shut down if deemed necessary. The selected compressor may be shut down to reduce energy consumption and greenhouse gas emissions such as carbon dioxide. The monitoring system (120) is configured to determine whether the cooling duty is greater than a cooling threshold.

[0064] In some embodiments, if the production train (112) has more than one compressor (e.g., a primary compressor and a secondary compressor), each compressor having a recycle valve (e.g., a primary recycle valve and a secondary recycle valve, respectively) then the control method may include closing the second recycle valve (111) of the secondary compressor of the production train (112) having the selected primary compressor. The production control system (107) may be configured to close the second recycle valve (111) of the secondary compressor of the production train (112) having the selected primary compressor.

[0065] In some embodiments, the control method (300) may include shutting down the secondary compressor of the production train (112) having the selected compressor (e.g., primary compressor). The production control system (107) may be configured to shut down the secondary compressor. The control method (300) may include deselecting the selected the compressor (e.g., primary compressor) if the corresponding production train (112) includes another compressor (e.g.,

secondary compressor) that is previously or is going to be shut down. The secondary compressor may have been shut down or is going to be shut down for, but is not limited to, maintenance, malfunctions, and/or poor efficiency. The monitoring system (120) may be configured to deselect the selected compressor (e.g., the primary compressor).

[0066] In some embodiments, the control method (300) may include selecting the respective compressor (e.g., the primary compressor of another production train) with a next lowest isentropic efficiency value. The monitoring system (120) may be configured to select the respective compressor with the next lowest isentropic efficiency value. The control method (300) may include opening the recycle valve (e.g., the first recycle valve (110)) of the selected compressor with the next lowest isentropic efficiency value. The control method (300) may include shutting down the selected compressor with the next lowest isentropic efficiency value. The production control system (107) may be configured to shut down the selected compressor with the next lowest isentropic value.

[0067] An example of the computer system (400) is described with reference to FIG. 4, in accordance with one or more embodiments. FIG. 4 is a block diagram of a computer system (400) used to provide computational functionalities associated with described algorithms, methods, functions, processes, flows, and procedures as described in the instant disclosure, according to an implementation. The illustrated computer (402) in the computer system (400) is intended to encompass any computing device such as a server, desktop computer, laptop/notebook computer, wireless data port, smart phone, personal data assistant (PDA), tablet computing device, one or more processors within these devices, or any other suitable processing device, including both physical or virtual instances (or both) of the computing device. Additionally, the computer (402) may include an input device, such as a keypad, keyboard, touch screen, or other device that can accept user information, and an output device that conveys information associated with the operation of the computer (402), including digital data, visual, or audio information (or a combination of information), or a GUI.

[0068] The computer (402) can serve in a role as a client, network component, a server, a database or other persistency, or any other component (or a combination of roles) of the computer system (400) for performing the subject matter described in the instant disclosure. The illustrated computer (402) is communicably coupled with a network (430). In some implementations, one or more components of the computer (402) may be configured to operate within environments, including cloud-computing-based, local, global, or other environment (or a combination of environments).

[0069] At a high level, the computer (402) is an electronic computing device operable to receive, transmit, process, store, or manage data and information associated with the described subject matter. According to some implementations, the computer (402) may also include or be communicably coupled with an application server, e-mail server, web server, caching server, streaming data server, business intelligence (BI) server, or other server (or a combination of servers).

[0070] The computer (402) can receive requests over network (430) from a client application (for example, executing on another computer (402)) and responding to the received requests by processing the said requests in an appropriate software application. In addition, requests may also be sent to the computer (402) from internal users (for example, from a command console or by other appropriate access method), external or third-parties, other automated applications, as well as any other appropriate entities, individuals, systems, or computers.

[0071] Each of the components of the computer (402) can communicate using a system bus (403). In some implementations, any, or all of the components of the computer (402), both hardware or software (or a combination of hardware and software), may interface with each other or the interface (404) (or a combination of both) over the system bus (403) using an application programming interface (API) (412) or a service layer (413) (or a combination of the API (412) and service layer (413)). The API (412) may include specifications for routines, data structures, and

object classes. The API (412) may be either computer-language independent or dependent and refer to a complete interface, a single function, or even a set of APIs. The service layer (413) provides software services to the computer (402) or other components (whether or not illustrated) that are communicably coupled to the computer (402). The functionality of the computer (402) may be accessible for all service consumers using this service layer (413). Software services, such as those provided by the service layer (413), provide reusable, defined business functionalities through a defined interface. For example, the interface may be software written in JAVA, C++, Python, or other suitable language providing data in extensible markup language (XML) format or another suitable format. While illustrated as an integrated component of the computer (402), alternative implementations may illustrate the API (412) or the service layer (413) as stand-alone components in relation to other components of the computer (402) or other components (whether or not illustrated) that are communicably coupled to the computer (402). Moreover, any or all parts of the API (412) or the service layer (413) may be implemented as child or sub-modules of another software module, enterprise application, or hardware module without departing from the scope of this disclosure.

[0072] The computer (402) includes an interface (404). Although illustrated as a single interface (404) in FIG. 4, two or more interfaces (404) may be used according to particular needs, desires, or particular implementations of the computer (402). The interface (404) is used by the computer (402) for communicating with other systems in a distributed environment that may be connected to the network (430). Generally, the interface (404) includes logic encoded in software or hardware (or a combination of software and hardware) and operable to communicate with the network (430). More specifically, the interface (404) may include software supporting one or more communication protocols associated with communications such that the network (430) or interface's hardware is operable to communicate physical signals within and outside of the illustrated computer (402).

[0073] The computer (402) includes at least one computer processor (405). Although illustrated as a single computer processor (405) in FIG. 4, two or more processors may be used according to particular needs, desires, or particular implementations of the computer (402). Generally, the computer processor (405) executes instructions (190) and manipulates data to perform the operations of the computer (402) and any algorithms, methods, functions, processes, flows, and procedures as described in the instant disclosure.

[0074] The computer (402) also includes a memory (406) that holds data for the computer (402) or other components (or a combination of both) that may be connected to the network (430). For example, memory (406) can be a database storing data consistent with this disclosure. In one example, memory (406) may store programs or algorithms for controlling operation of the control station (140) or the plurality of autonomous robots described above in accordance with one or more embodiments with reference to FIGS. 1-3. For example, the programs or algorithms may control operation of the control systems described with reference to FIG. 1. Although illustrated as a single memory (406) in FIG. 4, two or more memories may be used according to particular needs, desires, or particular implementations of the computer (402) and the described functionality. While memory (406) is illustrated as an integral component of the computer (402), in alternative implementations, memory (406) can be external to the computer (402).

[0075] The application (407) is an algorithmic software engine providing functionality according to particular needs, desires, or particular implementations of the computer (402), particularly with respect to functionality described in this disclosure. For example, the application (407) can serve as one or more components, modules, applications, etc. In one example, the application (407) may include programs or algorithms for controlling operation of the control station (140) or each of the plurality of autonomous robots described above in accordance with one or more embodiments with reference to FIGS. 1-3. For example, the programs or algorithms may control operation of control systems described with reference to FIG. 1 in accordance with one or more embodiments. Further, although illustrated as a single application (407), the application (407) may be implemented as

multiple applications (407) on the computer (402). In addition, although illustrated as integral to the computer (402), in alternative implementations, the application (407) can be external to the computer (402). In one example, the method described with reference to FIG. 4 may be implemented by the application (407).

[0076] There may be any number of computers (402) associated with, or external to, the computer system (400) containing computer (402), each computer (402) communicating over network (430). Further, the term “client,” “user,” and other appropriate terminology may be used interchangeably as appropriate without departing from the scope of this disclosure. Moreover, this disclosure contemplates that many users may use one computer (402), or that one user may use multiple computers (402). Furthermore, in one or more embodiments, the computer (402) is a non-transitory computer readable medium (CRM).

[0077] Embodiments of the present disclosure may provide at least one of the following advantages. The control method and associated systems may provide agility and flexibility in determining the power consumption status of various compressors as well as and implement remediation operations to alleviate future power consumption issues and carbon emissions. The control method may include optimizing energy consumption of the refrigeration system (106) where the energy consumption may reduce carbon dioxide emissions. The monitoring system (120) is configured to optimize energy consumption of the refrigeration system (106).

[0078] Although only a few example embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims.

Claims

1. A method for monitoring and controlling energy consumption of a refrigeration system operatively connected to a plurality of production trains, the method comprising: using a monitoring system, initiating a control logic configured to control a production control system pertaining to the refrigeration system, wherein the plurality of production trains includes a production flow therethrough, wherein each of the production trains respectively comprises a primary compressor comprising a first recycle valve; for each production train: obtaining a plurality of production parameters, wherein one of the plurality of production parameters comprises a production flow rate; determining whether the production flow rate is less than a production threshold; calculating, using the plurality of production parameters, an isentropic efficiency value for the primary compressor if the production flow rate is less than a production threshold; comparing each of the plurality of isentropic efficiency values; determining from the comparing each of the plurality of isentropic efficiency values, a primary compressor having a lowest isentropic efficiency value, wherein the lowest isentropic efficiency value is one of the plurality of isentropic efficiency values that is quantitatively smallest compared to all other isentropic efficiency values; selecting the respective primary compressor having the lowest isentropic efficiency value; and using the production control system: reducing the production flow to the one of the plurality of production trains which includes the selected primary compressor; redirecting the production flow from the one of the plurality of production trains including the selected primary compressor to at least one other of the plurality of production trains; opening the first recycle valve of the selected primary compressor to fully opened; and shutting down the selected primary compressor.

2. The method of claim 1, wherein the plurality of production parameters further comprises: a recycle valve opening measurement, and a driver power value; calculating, using the recycle valve opening measurement and the driver power value, the isentropic efficiency value.

3. The method of claim 2, further comprises: determining the driver power value, using the

- monitoring system, from at least a measured voltage, a measured current, and a power factor.
- 4.** The method of claim 1, further comprises: wherein each of the plurality of production trains comprises a secondary compressor comprising a second recycle valve, closing the second recycle valve of the secondary compressor of the production train having the selected primary compressor.
 - 5.** The method of claim 4, further comprises: shutting down, using the production control system, the secondary compressor of the production train having the selected primary compressor; using the monitoring system, deselecting the primary compressor having the lowest isentropic for shutdown if the secondary compressor of the production train having the selected primary compressor is already shutdown, selecting the respective primary compressor with next lowest isentropic efficiency value; shutting down, using the production control system, the selected primary compressor.
 - 6.** The method of claim 1, further comprises: wherein the monitoring system comprises a user interface, initiating, using the monitoring system, the control logic from input from the user interface.
 - 7.** The method of claim 1 wherein the production threshold is 1450 million standard cubic feet per day (mmscfd).
 - 8.** The method of claim 1, further comprises, using the monitoring system: calculating a cooling duty of the refrigeration system for liquefying the production fluids; determining whether the cooling duty is greater than a cooling threshold; deselecting the selected primary compressor.
 - 9.** The method of claim 1, further comprises: optimizing energy consumption of the refrigeration system, wherein optimizing energy consumption reduces carbon dioxide emissions.
 - 10.** The method of claim 1, further comprises: redistributing production flow equally among the plurality of production trains that are not the production train having the lowest isentropic value.
 - 11.** A system for monitoring and controlling energy consumption of a refrigeration system operatively connected to a plurality of production trains, the system comprising: a monitoring system configured to: initiate a control logic configured to control a production control system pertaining to the refrigeration system, wherein the plurality of production trains includes a production flow therethrough, wherein each of the production trains respectively comprises a primary compressor comprising a first recycle valve; for each production train: obtain a plurality of production parameters, wherein one of the plurality of production parameters comprises a production flow rate; determine whether the production flow rate is less than a production threshold; calculate, using the plurality of production parameters, an isentropic efficiency value for the primary compressor if the production flow rate is less than a production threshold; compare each of the plurality of isentropic efficiency values; determine from the comparing each of the plurality of isentropic efficiency values, a primary compressor having a lowest isentropic efficiency value, wherein the lowest isentropic efficiency value is one of the plurality of isentropic efficiency values that is quantitatively smallest compared to all other isentropic efficiency values; select the respective primary compressor having the lowest isentropic efficiency value; and the production control system configured to: reduce the production flow to the one of the plurality of production trains which includes the selected primary compressor; redirect the production flow from the one of the plurality of production trains including the selected primary compressor to at least one other of the plurality of production trains; open the first recycle valve of the selected primary compressor to fully opened; and shut down the selected primary compressor.
 - 12.** The system of claim 11, wherein the plurality of production parameters further comprises: a recycle valve opening measurement, and a driver power value; wherein the monitoring system is configured to: calculate, using the recycle valve opening measurement and the driver power value, the isentropic efficiency value.
 - 13.** The system of claim 12, wherein the monitoring system is configured to determine the driver power value from at least a measured voltage, a measured current, and a power factor.
 - 14.** The system of claim 11, wherein each of the plurality of production trains comprises a

secondary compressor comprising a second recycle valve, wherein the production control system is configured to close the second recycle valve of the secondary compressor of the production train having the selected primary compressor.

15. The system of claim 14, wherein the production control system is configured to: shut down the secondary compressor of the production train having the selected primary compressor; wherein the monitoring system is configured to: deselect the primary compressor having the lowest isentropic for shutdown if the secondary compressor of the production train having the selected primary compressor is already shutdown, select the respective primary compressor with next lowest isentropic efficiency value; wherein the production control system is further configured to shut down the selected primary compressor.

16. The method of claim 11, wherein the monitoring system comprises a user interface, wherein the monitoring system is configured to initiate the control logic from input from the user interface.

17. The system of claim 11 wherein the production threshold is 1450 million standard cubic feet per day (mmscfd).

18. The system of claim 11, wherein the monitoring system is configured to: calculate a cooling duty of the refrigeration system for liquefying the production fluids; determine whether the cooling duty is greater than a cooling threshold; deselect the selected primary compressor.

19. The system of claim 11, wherein the monitoring system is configured to optimize energy consumption of the refrigeration system, wherein optimizing energy consumption reduces carbon dioxide emissions.

20. The method of claim 1, wherein the production control system is configured to redistribute production flow equally among the plurality of production trains that are not the production train having the lowest isentropic value.
